

Pixel Permutation with Chaos: Secure Image Encryption using 2D Chaotic Function and DCT Compression

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Abstract— The proposed technique in this project involves compressing and encrypting images using a 2-dimensional chaotic function. The process begins by applying the Discrete Cosine Transform (DCT) to the input image to obtain its sparse representation. The DCT coefficients are then compressed by multiplying it with a Bernoulli matrix to obtain a compressed image and a 2D chaotic function is used to generate a chaotic sequence which is then sorted to obtain permutation indices. The permutation indices are used to rearrange the pixels in the image to create an encrypted image. The decryption of the project starts with generating the permutation indices which were used for encryption using the proposed 2D chaotic map, the initial parameters used for encryption will be transmitted through a secure channel, these permutation indices will be used to rearrange the pixels in their original location and the next decryption step, decompressing the image which was compressed using the Bernoulli matrix will be done using the SL01 recovery algorithm.

Keywords - image compression, image encryption, chaotic function, Discrete Cosine Transform (DCT), sparse representation, Bernoulli matrix, chaotic sequence, permutation indices, pixel rearrangement

I. INTRODUCTION

In the current era of technology, where images are frequently shared and stored in large quantities, the need for security and privacy is very high. It is important to have privacy and provide security while sharing resources like images, where interruptions are common. One way to achieve this is by encrypting the images using established methods like the Advanced Encryption Standard (AES) or encryption techniques. However, these methods are computationally intensive and may not be practical for real-world applications.

To address this challenge, this research aims to simplify the encryption process while ensuring reliability. We propose a method that combines Compressive Sensing with a novel 2-dimensional chaotic map to enhance the security of images while also addressing challenges associated with complexity and key management. By using these techniques, the goal is to provide a versatile and effective solution that not only safeguards image data from unauthorized access but also ensures a straightforward and adaptable implementation.

As this research progresses, we will discuss the details of our proposed method, present experimental results to validate its theoretical proposal, compare its advantages over existing approaches, and outline potential applications.

II. RELATED WORKS

In this decade, image encryption has gained significant attention because of the increase in demand for secure transmission of data and storage. Many encryption techniques are used widely to improve security and storage. Compressive sensing (CS) [1,2] reduces the amount of data that is going to be encrypted by reducing the size of the image. CS generates

a sparse image from which the original image could be reconstructed. The compressed sensing technique does not promise to generate the original image with every feature and pixel in same values, but it ensures to preserve the similarity of the image more than 75% with the original image. Most natural images are sparse by nature which makes CS faster to implement without extensively selecting pixels for sparse image.

A number of research works have used the integration of CS and chaotic maps for image encryption [3,4]. Chaotic maps are deterministic systems that shows unpredictable behavior, which makes them suitable for generating encryption keys. By combining the advantages of CS and chaotic maps, these methods achieve security and efficiency.

A Research work in this area by Zhang et al. [5], an encryption algorithm for image using CS and a novel chaotic function. The authors used their own chaotic map to generate a random projection matrix for CS, which was used to encrypt the image. This methodology proposed by Li et al. [6] introduced. The authors used CS and the measurement matrix required to perform CS was generated by the usage of a chaotic logistic map. This encryption algorithm proposed by them was robust and reliable.

Previous researches have used the integration of compressive sensing and chaotic systems for secure image encryption. The work proposed by scholars in [7] divided (decomposed) the input image into two components using the discrete wavelet transform (DWT). Encrypted using a novel chaotic function and compressed sensing. Reconstruction of the image was performed using the compressed sensing reconstruction algorithm.

The work presented by authors in [8] was about image encryption in cloud. This algorithm used a chaotic system for generating a secret key, which was then used to encrypt the compressed sensing measurements. The encrypted measurements were then stored in the cloud. In case of data loss, the image signal can be recovered using the secret key and the cloud-stored measurements.

Authors in [9] have encrypted the images using secured user name and compressive sensing.

Image encryption based on steganography techniques and chaotic maps were popular in recent years as well [10-13].

a image encryption algorithm that uses CS and a new chaotic function. The scheme uses a more efficient measurement matrix construction method and a more secure chaotic map, resulting in improved encryption performance and enhanced resistance to various attacks.

III. PROPOSED METHODOLOGY

Our proposed methodology involves compressing and encryption of a picture using chaotic function. First step commences by applying Discrete Cosine Transform to the image to be compressed and encrypted. The reason for choosing DCT is because it is capable of providing an efficient sparse representation [14,15] of the image, which provides a foundation for subsequent encryption steps. Following the DCT, the sparse image is multiplied with a bernoulli matrix. This step achieving compression. The dimension of random binary matrix is lesser than the sparser image and this size determines the amount of data that is actually compressed. The compressed images are then encrypted using a proposed 2-dimensional chaotic map. At the receiver side, the compressively sensed and encrypted image is recovered using Sl_0 recovery algorithm. The combination of Discrete Cosine Transform, Compressive sensing and 2d chaotic map provides a robust image encryption algorithm.

1) Proposed Chaotic Map for Enhanced Image Encryption

Chaotic maps are deterministic systems and they behave in an unpredictable and random manner, making them ideal for use in cryptography applications. a new chaotic function that was inspired from Tinkerbell map is proposed.

Chaotic function used in the project is mathematically represented as follows:

$$\begin{aligned} x_{n+1} &= x_n^4 - y_n^2 + ax_n + by_n \\ y_{n+1} &= 2x_n y_n + cx_n + dy_n \end{aligned} \quad (1)$$

where (x_n, y_n) represents the state of the system at time step n , and (x_{n+1}, y_{n+1}) represents the state of the system at the next time step $n+1$.

The parameters a , b , c , and d in the Tinkerbell map control the behavior of the system and determine the characteristics of its attractors.

Here's a brief description of each parameter:

- **a**: Influences the behavior of the x variable.
- **b**: Influences the interaction between the x and y variables.
- **c**: Influences the behavior of the x variable and the coupling between the x and y variables.

- **d**: Influences the behavior of the y variable.

The values of these parameters can be adjusted to explore different dynamical behaviors of the system, ranging from regular periodic motion to complex chaotic behavior.

2) Encryption Algorithm

The idea of image encryption which can also be called as pixel alteration algorithm using a unique chaotic function is represented in Fig. 1. The steps are detailed as follows.

i. Image Acquisition and Preprocessing

Acquiring the input image. Converting the image to grayscale if it is a colored image.

ii. Discrete Cosine Transform (DCT)

Apply DCT to the grayscale image to obtain the DCT coefficients for getting its sparse representation.

iii. Compressive Sensing (CS)

Multiplying the DCT coefficients with matrix generated to perform compression.

iv. Chaotic Map-Based Encryption

The 2-dimensional chaotic function is used to generate an arbitrary sequence of numbers equal to the number of pixels in the image and the sequence is sorted to create permutation indices, the pixels of the image are then rearranged in the order of their respective permutation indices.

3) Decryption

The compressed and encrypted image is sent to the receiver for recovery. The encrypted image is reconstructed using the four parameters a , b , c , d . The receiver uses the four parameters to create the original permutation indices to place the pixels back in their original place. Then, the SL_0 algorithm is used to reconstruct the compressed image.

Figure 1 shows the images of successful and unsuccessful decryptions, image a was decrypted using the original parameters used for encryption and image b was decrypted with a small variation in a parameter showing the robustness of the encryption algorithm.

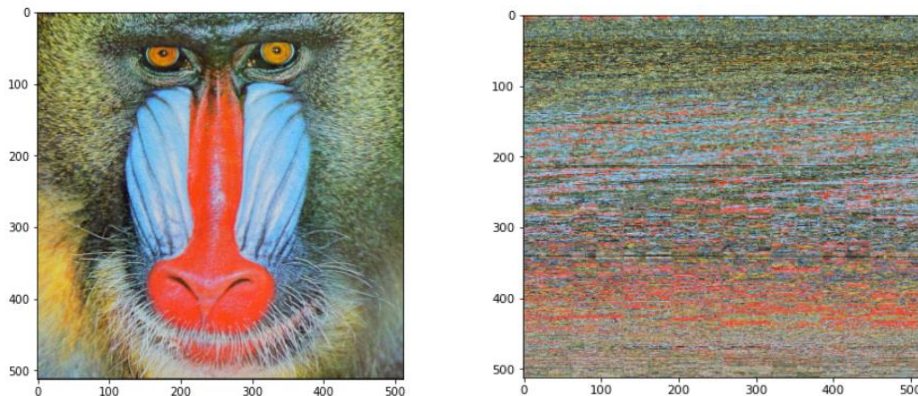


Fig.1 Image A) decryption successful and Image B) decryption failure

IV. RESULTS AND DISCUSSION

I. PSNR, SSIM and Correlation Analysis

In our project, we use PSNR and SSIM to check the quality of our images after compressive sensing. The **table 1** below shows the PSNR and SSIM values of compressive sensing performed on images using a Random Binary Matrix, Bernoulli Matrix and Gaussian Matrix.

PSNR that tells us how much our pictures change when we compress them, SSIM is another number, but it's different. It checks the details in our pictures and compares them to see if they look similar or not. If the PSNR and SSIM values are higher it shows the quality of images even after Compressive

Sensing. The Bernoulli Matrix gave the best results and so it is used for compressive sensing in this project.

In our research, we conducted correlation analysis to explore the relationship images before and after the encryption.

We check horizontal, vertical and diagonal correlation of images before and after encryption showing us how much the image after encryption is correlated horizontally, vertically and diagonally to the original images. The scatter plot of the correlation analysis represents the analysis visually. **Table 2** and **figure 2** shows the correlation analysis conducted.

Image	Random Binary Matrix		Bernoulli Matrix		Gaussian Matrix	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Peppers	37.0473	0.8984	38.0231	0.8432	35.6352	0.8043
Mandril	29.1021	0.8673	37.0642	0.8021	33.0531	0.7986
Pirate	31.5678	0.8226	38.5438	0.8678	34.2373	0.8451

Table 1 Evaluation metrics of Compressive Sensing

Image	Plain Image Correlation			Encrypted Image Correlation		
	Horizontal	Vertical	Diagonal	Horizontal	Vertical	Diagonal
Baboon	0.9046	0.8483	0.8181	0.0322	0.0316	0.0320
Fruits	0.9796	0.9795	0.9651	0.2057	0.2046	0.2049
Pirate	0.9593	0.9675	0.9412	0.0024	-0.0010	0.0001

Table 2 Corellation Analysis between plain image and encrypted image

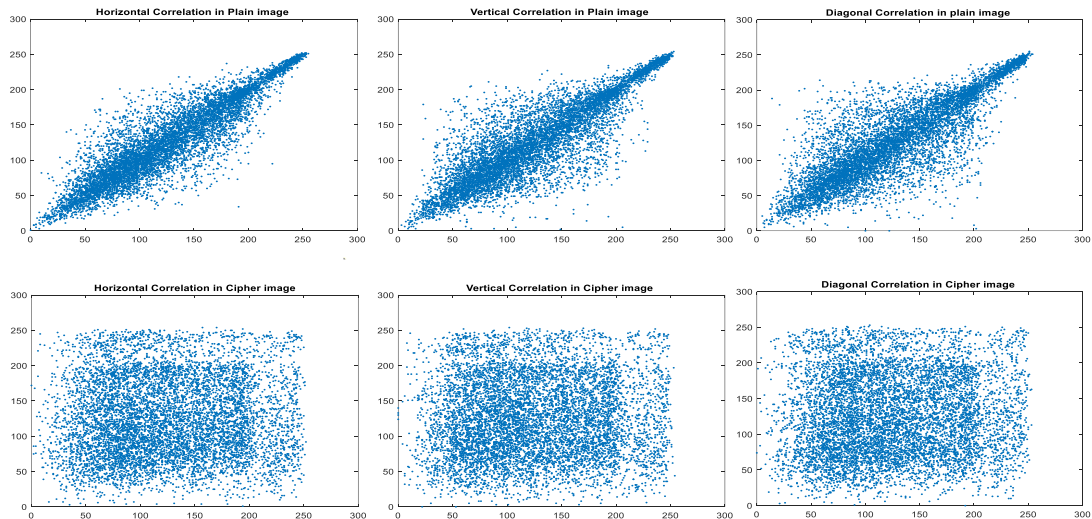


Fig.2 Visual Representation of Correlation Analysis

II. Histogram Analysis

In our research, we performed histogram analysis on both the original and compressed images to assess the impact of compression on pixel intensity distribution. We utilized MATLAB to generate histograms for each image, plotting the frequency of pixel intensity values ranging from 0 to 255. The histograms of the original and compressed images

revealed differences in pixel intensity distribution. While the original image exhibited a wide range of distribution of pixel intensities across the entire range, the compressed image histogram displayed uniform distribution showing the characteristics of a compressed image is uniform in pixel intensity distribution. **Fig 3,4** are the histogram plots of original and compressed images of a baboon and fruits.

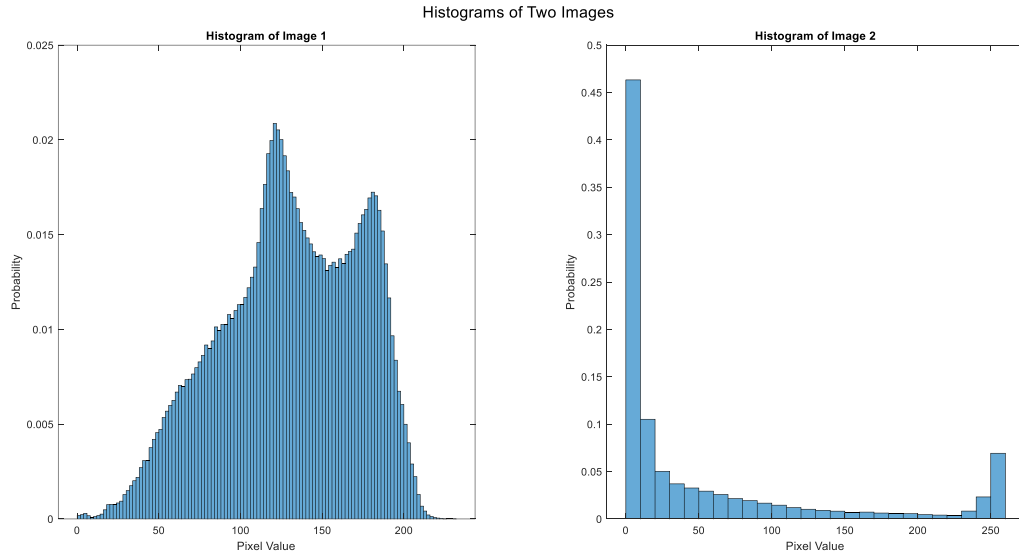


Fig.3 Histogram plots of original and compressed image of baboon.

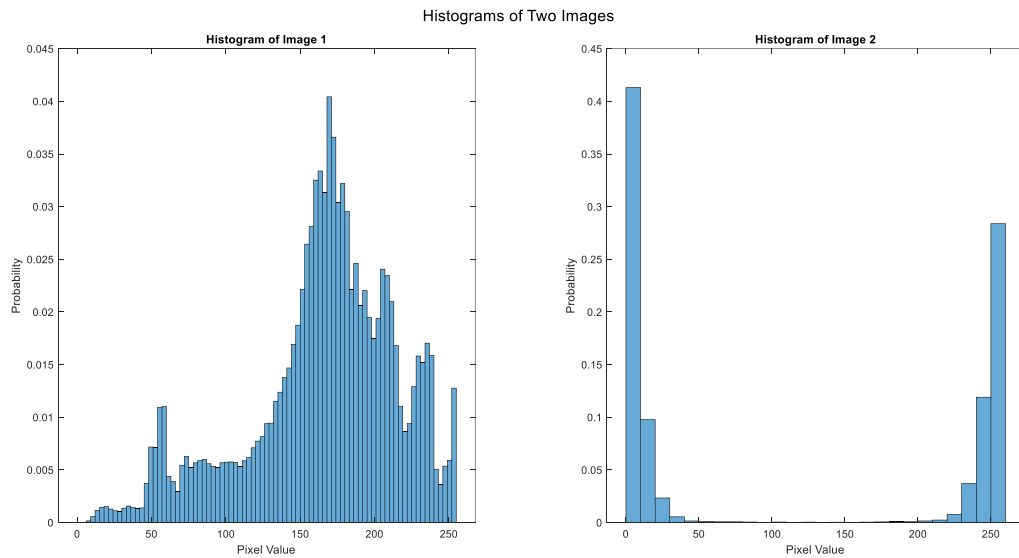


Fig.4 Histogram plots of original and compressed image of fruits.

Figure 5 represents the originl,encrypted and decrypted images of fruits,baboon and pirate ,the encrypted images representing the colour pattern of their original images shows the rearrangement performed using chaotic function.

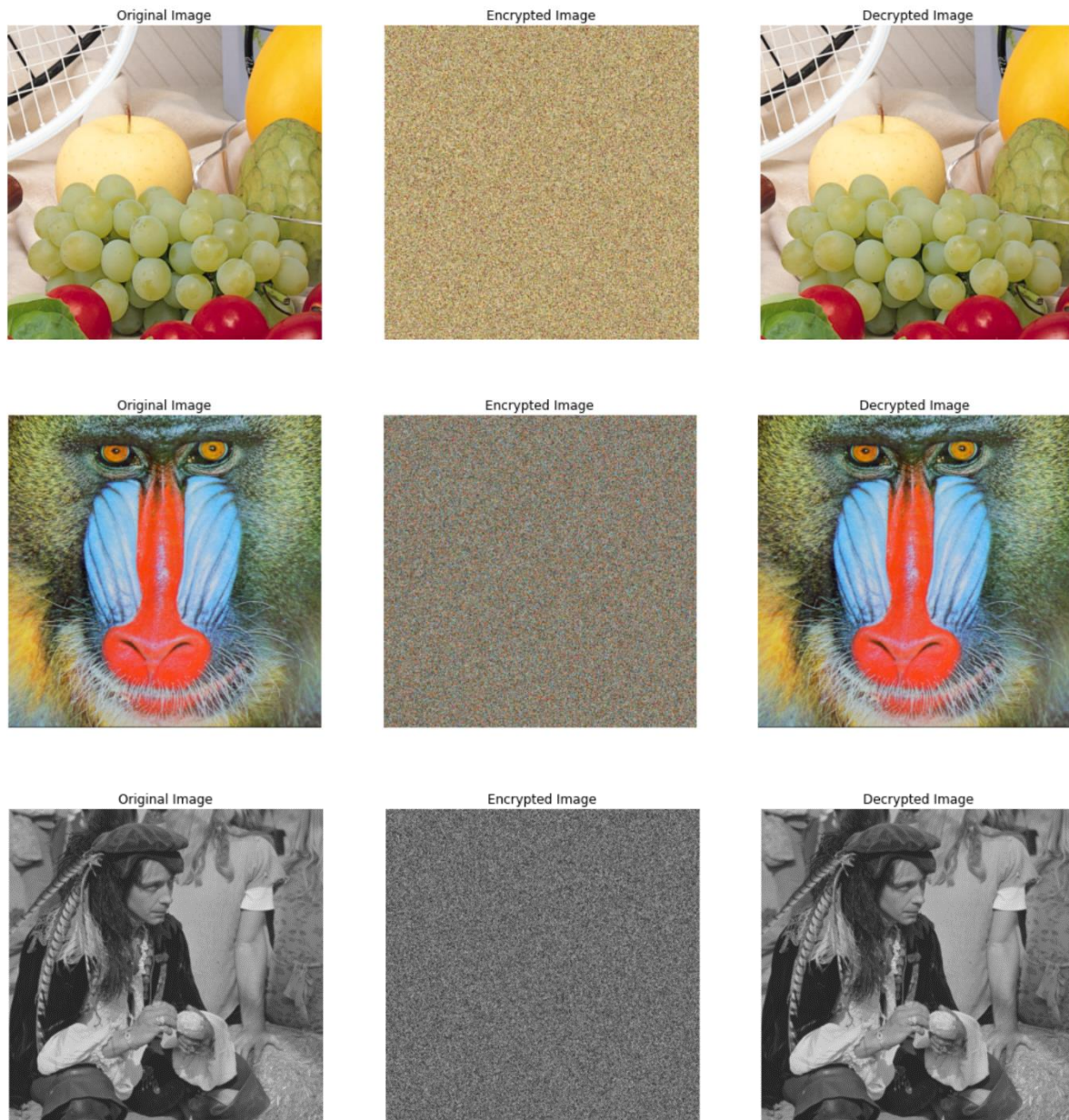


Fig.5 Original, encrypted and recovered images of (a) peppers, (b) mandril, (c) pirate

V. CONCLUSION

In this project, an image encryption algorithm is proposed. A 2-dimensional chaotic function with high randomness and a novel compression encryption algorithm is proposed. Experimental results prove that the proposed algorithm effectively compresses and encrypts digital images while preserving their visual quality in the decompression process. Overall, the proposed image encryption algorithm provides a secure and robust solution for protecting digital images from unauthorized access and malicious attacks. The combination of compressive sensing and the novel chaotic map ensures

efficient and effective image compression-encryption process, making it capable for implementing in real life.

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